Week 4 - 29/06/2026 Monday

My React Project - JavaScript vs TypeScript

Phase 1

Introduction

Root em and pixels are commonly used units in web design to measure the font size. Although it can seem like both measuring units provide the same output, they're actually very different in terms of web accessibility, and each has its own key points, and preferred approaches.

Pixels are considered absolute units, meaning that 1px == 1px. This inflexibility limits the web accessibility in terms of responsive design, meaning that it doesn't adapt to the user's preferences.

On the other hand, root em is adept and is very flexible when it comes to responsiveness in web accessibility and even small actions such as zooming in.
1rem == 16px.

Part 1. Components, Props, & States:

Components, props and states are the foundational building blocks of any React project**:**

1. Components: reusable and independent UI building blocks.
2. Props: fixed (immutable) data passed from parent to child.
3. State: unfixed (mutable) data managed within a component that triggers UI rerenders.

Functional components are the basic building blocks in React used to create and manage UI elements.
Props to pass data between components properties similar to function arguments in JavaScript and attributes in html. (optional props are those not required to return).
Example:

(https://www.w3schools.com/react/showreact.asp?filename=demo_react_props)

useState hook is a built-in React function that lets you add and manage local state within functional components. It tracks data changes across component re-renders.
When a user updates a useState variable, React automatically updates the UI accordingly.

Event handling and conditional rendering are the two foundational pillars that turn static React components into dynamic, interactive user interfaces.
The key difference:

1. Event handling captures the user's intent.
2. Conditional Rendering updates the visual display based on that intent.

The key concept is **Component Reusability**, which is a software development practice where distinct, self-contained (independent) pieces of code and UI elements are built to be used across multiple applications or various parts of the same project.
This practice has many benefits, including:

1. Reduces redundancy.
2. Accelerates development.
3. Ensures that the design and behavior are consistent.

Part 2. Lists, Forms, & Events:

Lists, forms, and events are the building blocks of interactive web applications in JavaScript.
In JavaScript, Lists are known as Arrays, and they store ordered collection of data.
Events are actions that happen in a browser during user navigation, which includes mouse clicks, submitting forms, pressing on links, etc. (ex: element.addEventListener('event_type', callback_function);).

Rendering lists with .map() means taking an array of raw data and transforming it into a collection of visual UI elements.

This saves the developers the trouble of having to rewrite the code manually for each item, and instead do it once and call it multiple times.
Example:
The process of doing so is quite simple. Let's say we have this list

```
<ul>
 <li>Creola Katherine Johnson: mathematician</li>
 <li>Mario José Molina-Pasquel Henríquez: chemist</li>
 <li>Mohammad Abdus Salam: physicist</li>
 <li>Percy Lavon Julian: chemist</li>
 <li>Subrahmanyan Chandrasekhar: astrophysicist</li>
</ul>
```

First thing we'll do is put them in an array:

```
const people = [
 'Creola Katherine Johnson: mathematician',
 'Mario José Molina-Pasquel Henríquez: chemist',
 'Mohammad Abdus Salam: physicist',
 'Percy Lavon Julian: chemist',
 'Subrahmanyan Chandrasekhar: astrophysicist'
];
```

Then, we're going to map the people into a new array of JSX nodes, listItems:

```
const listItems = people.map(person => <li>{person}</li>);
```

After mapping these specific members, we will return listItems from the component wrapped in between :

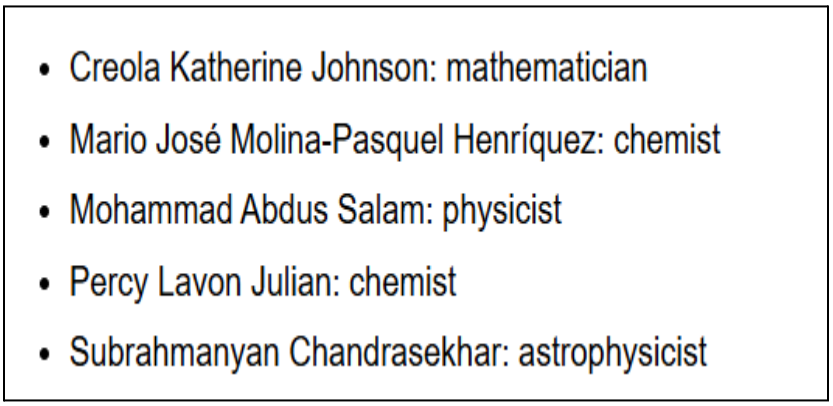
```
return <ul>{listItems}</ul>;
```

This is the outcome of the process:

App.js

```
const people = [
  'Creola Katherine Johnson: mathematician',
  'Mario José Molina-Pasquel Henríquez: chemist',
  'Mohammad Abdus Salam: physicist',
  'Percy Lavon Julian: chemist',
  'Subrahmanyan Chandrasekhar: astrophysicist'
];

export default function List() {
  const listItems = people.map(person =>
    <li>{person}</li>
  );
  return <ul>{listItems}</ul>;
}
```



(https://react.dev/learn/rendering-lists)

Keys and Controlled Components are core concepts in React that manage how elements maintain their identity, preserve state, and sync data between the UI and logic of JavaScript.

Form submission handling is the process of capturing a user's input and transmitting it to a server or API. This process includes intercepting, validating, and processing data entered by a user into an HTML form.

Input validation and multiple input handling:

1. Input validation: checks if data is safe, in the correct format, and meets specific criteria, such as password lengths and email patterns.
2. Multiple input handling: the structural mechanism that tracks, updates, and processes data from several different fields simultaneously on the UI.

Basically, input validation verifies that the data provided by the user is safe and meets the expectations or specifications. On the other hand, multiple input handling manages how applications capture and store data for multiple fields at once.

Part 3. Implementation

After reviewing the key points and my JavaScript vs TypeScript project, I will document below where the points are implemented:

First Draft:

Concept	Status
Functional Components	Applied
Props (properties)	Applied
useState hook	Applied
Event Handling	Applied
Conditional Rendering	Applied
Component Reusability	Applied
Rendering lists with .map()	Applied
Keys in React	Applied
Form Submission Handling	Missing
Single & Multiple Input Validation	Missing

TO DO:

1. Convert px to rem (font size) [DONE]
2. Use padding to remove the space around the centered dashboard [DONE]
3. Components: individual and shared ones => header, footer, and navbars [DONE]
4. Divide these into different components [DONE]
5. Deep understanding of the report + code + files [DONE]

Phase 2

Check list for phase 2:

useEffect & API Integration

- useEffect hook (lifecycle, side effects) [DONE]
- Fetching data from APIs, Loading states & Error handling [DONE]
- Critical: Connecting React to their WebAPI backend [DONE]
- CORS basics and how to handle it [DONE]

useEffect hook: used in React to handle side effects, which is anything a component needs to do that requires:

1. An external API.
2. Fetching data from a server.
3. Starting a timer.
4. Setting up a subscription.

In other words, it tells React to perform a specific action after rendering.

useEffect(() => {}) → after every render.

useEffect(() => {}, []) → only once after the first render.

useEffect(() => {}, [value]) → whenever value changes.

return () => {} inside useEffect → cleanup before unmounting or before the effect runs again.

CORS: Cross-origin resource sharing

Security feature in web browsers that restricts web pages from requesting data (ex: APIs) from a domain that's different from the one that served the original page.

How can we handle a CORS error?

This issue can be resolved by configuring the backend server and making it return the correct HTTP headers. These headers authorize the frontend domain to access its resources.

Week 4 - 01/07/2026 Wednesday

React Router - Part 1.1

React Router is a system that allows a react application to display different pages or views without reloading the entire website **[**Single Page Application**]**.

Reloading the entire website happens by default in normal applications. The process behind it is: user clicks a link → request is sent from browser to the server → returns a whole new html page → full page refresh is caused.

However, the process in React Router is different; the parent content remains fixed in the main page, and only the changeable content (child) is updated **[**No page refresh**]**.

Task:

- Installing React Router [DONE]Routes and navigation [DONE]
- Link components [DONE]
- Route parameters [DONE]
- Nested routes [DONE]
- Focus: Building multi-page applications [DONE]
- Practice : Add routing structure to previous project [DONE]

Week 4 - 02/07/2026 Thursday

React Router - Part 2

Task:

- Dynamic routes with parameters [DONE]
- Query parameters [DONE]
- Programmatic navigation (useNavigate) [DONE]
- Route guards/protected routes basics [DONE]
- 404 Not Found pages [DONE]
- Navigation state [DONE]
- Practice : Implement complete navigation system [DONE]

Req 1 - Dynamic route with parameters: flexible URL pattern that captures variable data (ID, slug) rather than relying on static hardcoded paths. Instead of hardcoding each page, [TopicView.tsx] and [LevelView.tsx] can both read the parameters in [App.tsx], and it renders any topic or level.

Req 2 - Query parameters [ArrivedViaSearchBanner.tsx, Ln13-14**]:** defined set of parameters attached to the end of a URL used to provide additional information to a web server when making requests.

Req 3 - Programmatic navigation [useNavigate**]:** the concept of useNavigate is making the website move on to the next level based upon completion ONLY.
Implementation:

1. **[**LevelView.tsx**]:** if the user answers a question correctly, only then can she/he unlock the next level.
2. **[**QuestComplete.tsx**]:** once the quest is completed, users can press “Play Again!”, and it’ll take them back to level 1.
3. **[**SearchBar.tsx**]:** calls navigate(...) to send them to their destination and attaches a tag in the URL that tells you the page title + a hidden note with the actual word the user typed. Finally, once the user is done, the search bar gets cleaned up. This follows a dynamic approach since the search is different every single time.

Req 4 - Route guards / protected routes: before moving on to the next page, it must check the conditions are met beforehand. I have implemented this in **[**RequiredLevelUnlocked.tsx**]** so the user cannot move on to the next level without clearing the current one.

Req 5 - 404 Not Found error [NotFound.tsx**]:** if the user types a URL that doesn’t match any of the routes that I have assigned, the error page will appear

Req 6 - Navigation state: a secret note that is sent along the navigation that doesn’t show up in the URL.

Param → part of the path that identifies what page we’re in.
Query → visible log on the URL - extra details, shareable, bookmarkable.
State → invisible data alongside the navigation - extra detail - refresh == gone.

Week 5 - 05/07/2026 Sunday

Context API & Global State

Task:

- When to use global state vs local state [DONE]
- Creating Context [DONE]
- useContext hook [DONE]
- Provider pattern [DONE]
- Multiple contexts [DONE]
- Use case: User authentication state, theme, app settings [DONE]
- Practice : Implement global user authentication state [DONE]

When to use global state vs local state

Name	Local State	Global State
Usage	Used for data that affects only one component	Used when multiple distant components need to share and synchronize the same data user authentication
Examples	Form inputs UI toggles Modals	Dark / Light theme settings Global caches Shopping cart data

Req 1 - Creating Context
Generates a container that can hold a value and be read anywhere in the component tree below it without the value being passed down as a prop through every level. It uses the prop drilling approach where the context skips the middlemen entirely, meaning that if the AccountPage needs user, it would have to pass multiple layers down to the AccountPage.

Req 2 - useContext [AuthContext.tsx, SettingsContext.tsx, ThemeContext.tsx]
Allows a functional component to read and subscribe to context from anywhere in the component tree without manually passing props down through intermediate components (solves prop drilling).

Req 3 - Provider Pattern
Also solves prop drilling issue as it shares data across a global or nested component tree without passing props manually through every level.
It also syncs to <localStorage> via <useEffect>, so login state, theme, and settings survive a page refresh.

Req 4 - Multiple Contexts [AppProviders.tsx]
That file nests all four providers → Auth, Settings, Theme, QuestProgress.
Q / Why do we split separate contexts instead of using one big context?
Avoids unnecessary re-rendering: if auth, theme, and settings were one context, then toggling the theme would re-render every component that only cares about auth data. Splitting them means a component calling useTheme() doesn't re-render when settings change.

Week 5 - 06/07/2026 Monday

Part 1 - Advanced State Patterns:

State patterns in React refer to the standard strategies, design architectures, and best practices developers use to structure, update, and share data within an application.

Task:

- **useReducer hook for complex state [DONE]**
- **Combining Context with useReducer [DONE]**
- **State management best practices [DONE]**
- **Avoiding prop drilling [DONE]**
- **When to lift state up [DONE]**

Practice: Refactor application to use better state management [DONE]

Part 2 - Axios & API Integration

Axios is a popular, promise-based JavaScript library used to send asynchronous HTTP requests. It works seamlessly in both web browsers and Node.js environments.

Task:

- **Installing and configuring Axios** [DONE]
- **Axios vs Fetch API** [DONE]
- **Creating Axios instances** [DONE]
- **Interceptors for adding auth headers** [DONE]
- **Error handling with Axios** [DONE]
- **Request/response transformation** [DONE]

Practice: Replace Fetch with Axios and add interceptors [DONE]

Part 1 - Advanced State Patterns:

Req 1 - useReducer for complex state

What it is: instead of useState, we describe state changes as small named actions & a single function (reducer) that computes the next state from the current state + action.

Built-in function used to manage complex state logic.

(useState is for light, simple projects. useReducer is for big, complex ones).

Code implementation [questProgressReducer.ts Ln13-33] then [QuestProgressContext.tsx Ln9]

Req 2 - Combining Context with useReducer

What it is: useReducer alone only manages state inside one component. Context makes state available to any component in the tree. We put them together when several unrelated components need the same reducer-managed state.

Reducer decides how the state changes. Context decides who can see it.

<cleared> (finished levels) is needed by 5 separate components. useReducer is inside one of them only and can't reach the other, so:

1. **Context comes in to share it.**
2. **useReducer keeps the updated logic clean once shared.**

We use both Context and useReducer together whenever state is complex and needed in multiple places.

Where it's implemented: [QuestProgressContext.tsx Ln7-24]

Reducer owns the logic behind how state changes.

Provider owns how it's exposed.

Req 3 - State management best practices

1. **Not to use useReducer everywhere. AuthContext.tsx and ThemeContext.tsx kept useState one simple value. No need to add reducer ceremony (Reducer ceremony is the structured, multi-step process required to use the useReducer hook).**
2. **Memoize the Context value → memoizing is giving memory to the code to prevent wasting time recalculating the exact same thing twice.**
3. **Keep reducers in their own file with no React code → this makes it easy for them to read and test on their own.**

Req 4 - Avoiding prop drilling

How I avoided prop drilling in my project:

1. **Quest state was drilled through the router instead of being used directly.**

[QuestLayout.tsx] was pulling quest data from Context, then repassing it down through React Router's outlet, & the three child routes [LevelView.tsx], [QuestComplete], and [RequiredLevelUnlocked.tsx] were digging it back out ⇒ This process == Prop Drilling.

Fix: removed the relay and had each child call the Context hook directly.

Req 5 - When to lift state up

Lift state up in React should be lifted when two or more components need access to the same changing data to ensure a single source of truth. By moving the state to their closest common ancestor, we allow data to flow down as props while updates pass up via callback functions.

Part 2 - Axios & API Integration

Req 1 - Installing and configuring Axios:

To install it I wrote in my terminal "npm install axios" then it got downloaded in my package.json dependencies.

Req 2 - axios vs fetch API:

Fetch and Axios are commonly used web applications to make HTTP requests from the frontend to backend.

Fetch → modern browsers.

Axios → simple library with extra features.

Fetch is a JavaScript feature available in modern browsers that uses the fetch() method to make network requests.

1. **Fetch() → make network requests and return a promise**
2. **then() → handle successful response**
3. **catch() → handles any errors that occur during the request**

Axios is a 3rd party JavaScript library designed to make HTTP requests, which works with both browsers and Node.js. Axios uses promise API introduced in ES6, and is known for its simplicity and extra features, such as:

1. **Request/response interception**
2. **Error handling**
3. **Support for cancellation**

——--

1. **axios.get() → make a GET request to the specified URL and return a promise**
2. **.then() → handle successful response**
3. **.catch() → handle any errors that occur during the request**

Fetch is used for simple, light projects without the need of additional libraries.

Axios is used for heavier, complicated projects that require advanced features and a higher level of convenience when handling requests.

Req 3 - Creating axios instances

Creating an Axios instance means instead of using the plain, global axios object everywhere, we make our own copy of it with axios.create({...}), pre-loaded with settings we don't want to repeat.

Req 4 - Interceptors for adding auth headers

An interceptor is code that automatically runs on every request or response that goes through an Axios instance.

User doesn't call it, axios calls it for us

1. **Right before the request goes out.**
2. **Right after the response comes back.**

Request interceptor for auth headers specifically means: automatically attaching our login token to outgoing requests, so we don't have to remember to add it manually every single time we call an API.

[httpClient.ts Ln 40-55] Code Breakdown:

1. **httpClient.interceptors.request.use(...) — "before you send any request through httpClient, run this function first."**
2. **config is the request that's about to be sent (its URL, headers, method, etc.) — this function gets a chance to inspect or modify it before it actually goes out.**
3. **localStorage.getItem(AUTH_STORAGE_KEY) — checks if someone is logged in, by reading the same key AuthContext uses to store the user.**
4. **If someone's logged in, config.headers.set("Authorization", "Bearer <username>") attaches an Authorization header — this is the standard way APIs check "who is making this request?"**
5. **return config — hands the (now-modified) request back to Axios so it can actually be sent.**

Req 5 - Error handling with Axios

httpClient.interceptors.response.use(...) catches every failure and turns it into one consistent ApiError. In my case, I handle ERROR 429 - Too many requests (user waits 1s) & Network failure

Req 6 - Request/response transformation

the same interceptor reads the Retry-After header on a 429 response and converts it into milliseconds my retry logic can actually use.

Practice: Replace Fetch with Axios and add interceptors

that's the whole api.ts rewrite- done.

Q/ Why did I use React Router DOM and not React Route?
I used react-router-dom instead of react-router because react-router-dom is specifically designed for web browsers.
It provides the exact components required to interact with the browser's URL and history API.

Week 5 - 07/07/2026 Tuesday

Authentication & Component Libraries

Authentication Implementation

• Login/logout functionality [DONE]
• Storing JWT tokens (localStorage vs sessionStorage) [DONE]
• Protected routes implementation [DONE]
• Auth context setup [DONE]
• Redirect after login [DONE]
• Token expiration handling [DONE]
Practice : Complete authentication system with their backend [DONE]

:User Session Management

• Persistent login (remember me) [DONE]
• Auto-logout on token expiration [DONE]
• Refresh token implementation (if backend supports) [DONE]
• User profile management [DONE]
• Authorization (roles and permissions) [DONE]
Practice : Enhance auth system with session persistence [DONE]

Part 1 - Authentication implementation

Req 1 - Login/logout functionality [contexts/AuthContext.tsx]

login is an async function; it sends the username/password to the backend and *waits* for a reply before doing anything else.
setIsLoading(true) flips a flag for the UI to read and disable the button then shows "Logging in..." while it waits.
If loginRequest throws (bad password, network error, whatever), the catch block turns that into a readable message and re-throws it, so the form component can also react if it wants to.
finally always turns isLoading back off, success or failure, so the button never gets stuck disabled.
-
logout is not async — notice there's no await on logoutRequest. It fires that request and immediately moves on to endSession(), which clears everything locally

.catch(() => {}) on the request means "if this network call fails, silently ignore it — don't let it block anything."

Why was it implemented this way?

Login has to be a slow, careful process because we don't know if the user is authenticated until the server says so, so the UI has to wait and show an error.

Logout is the opposite as it should be instant.

Req 2 - Storing JWT tokens (Json web tokens- verify the user's identity and permissions after logging in) [features/auth/session.ts]

localStorage and sessionStorage have the exact same API (.setItem, .getItem, .removeItem) the only difference is when the browser clears them → localStorage survives closing the tab/browser, sessionStorage doesn't.

Because they share an API, target can just be a variable pointing at whichever one applies, based on the persist boolean — no need to write the save logic twice.

other.removeItem(...) deletes any leftover copy in the *other* storage, so a token never silently exists in both places at once (which would make "am I remembered?" ambiguous later).

Req 3 - Protected routes implementation [auth/RequireAuth.tsx Ln13-28]

This is a wrapper component. It doesn't render AccountPage at all unless isAuthenticated is true.

<Navigate> in React Router's way of saying redirect the browser to this URL.

It is considered a component and not a function call because React Router needs it to happen as part of the rendering, not side effect.

<replace> means it swaps the current history entry instead of adding a new one, so hitting the browser's Back button after logging in doesn't take you back to the "blocked" state.

No rendering occurs when the user is logged out, it leaves the user stuck on a blank page.

Redirecting to /login route means the URL bar reflects what's happening and that the same pattern is used somewhere else in the same application. (same concept as the 3 levels and having to complete one to move on to the next.

Req 4 - Auth context setup

createContext(null) makes a "box" that can hold a value and be read from anywhere below it in the component tree, without passing it down through every component's props by hand.

AuthProvider is the component that fills that box (it holds the real session state and computes a value object to put in the box, and any value that calls useAuth() pulls the value back out).

Error handling: The if (!ctx) throw guard exists so that if a user ever accidentally uses useAuth() outside of <AuthProvider>, they get a clear error immediately instead of a confusing undefined.something crash somewhere else.

Req 5 - Redirect after login
[auth/RequireAuth.tsx] → stashes where you were headed (hides)
[auth/LoginPage.tsx] → reads it back
React Router lets you attach any data to a navigation via state- it's not visible in the URL, it just rides along with that specific navigation.
when RequireAuth bounces you to /login, it also hands over "here's the page you were actually trying to reach" (location, which includes the pathname).
Why do we use this instead of a URL?
A query string is visible and editable in the address bar + needs manual encoding if the path has special characters.
Router state is invisible, disappears naturally after use, and can't be tampered with by just editing the URL.

Req 6 - Token expiration handling [auth/jwt.ts]
Meaning: creating a system that detects when a user's security token (like a JWT) becomes invalid and seamlessly updates it or logs the user out without crashing the app or confusing the user.
JWT can look like: header.payload.signature 3 base64 encoded chunks separated by dots.
decodeJwt reads the middle chunk → middle chunk == JSON object from the backend.
Implementation notes:
JWT stores exp in seconds, but JavaScript's Date.now() works in milliseconds, which is why getExpirtyTime multiplies by 1000.
isTokenExpired is like saying "is right now past that time?"

Week 5 - 08/07/2026 Wednesday

Data Tables & Pagination

• Displaying large datasets efficiently [DONE]
• MUI Data Grid or Ant Design Table or React Table [DONE]
• Server-side pagination implementation [DONE]
• Sorting and filtering [DONE]
• Search functionality [DONE]
• Column customization [DONE]
Practice : Build data table connected to backend with pagination [DONE]

Req 1 - Displaying large datasets efficiently [features/library/DataTablePage]
Pagination meaning: a web development technique used to split a massive dataset into smaller, discrete chunks (pages) instead of rendering all data simultaneously.
The table doesn't render the whole dataset at once. Only the current page's rows (rows) are held in component state and passed to the grid. [Ln 58-59].
Why is rowCount tracked separately from rows.length?
-rowCount represents the total number of matching records across all pages.
-Rows only hold 1 page's worth of data.
→ This is what lets the grid show "1–10 of 20" style pagination controls without ever holding all 20+ rows in memory at once — the same pattern is what makes this scale to thousands of rows without a performance hit, since the browser only ever renders one page.

Req 2 - MUI Data Grid [features/library/DataTablePage Ln4-7]

```
import {
  DataGrid,
  type GridColDef,
  type GridPaginationModel,
  type GridSortModel
} from "@mui/x-data-grid";
```

(DataGrid) is the table component itself.
Why did we choose this method instead of a plain html <table>?
Using DataGrid from MUI's x-data-grid package comes with pagination, storing, and column resizing built in. It simplifies the dev's job; instead of implementing grid behavior from scratch, we only have to supply data and configuration.
MUI: Material-UI- is an open-source React component library that provides pre-designed, production-ready user interface (UI) elements. It drastically speeds up web development by allowing developers to import ready-made elements instead of coding CSS from scratch.
(GridColDef) types the column definitions.

(GridPaginationModel & GridSortModel) type the pagination and sorting state that get passed back and forth with the grid.

Req 3 - Server-side pagination implementation [features/library/DataTablePage.tsx Ln61 Ln144-160]

Ln61: paginationMode="server" tells the DataGrid "don't paginate the data yourself — I'll hand you exactly the rows for the current page, you just render them and show the page controls."

paginationMode="server": paginates the data and provides the rows for the current page.

DataGrid: renders them.

Ln152: Whenever the user clicks a page number or changes page size, onPaginationModelChange fires and updates paginationModel, which triggers a useEffect that refetches from adventurersAPI.ts

Q/Why did we choose server-side instead of client-side pagination?

with client-side pagination, the grid would need the *entire* dataset in the browser and would just hide/show rows locally. Server-side pagination instead only ever transmits and holds one page of data at a time, which is what makes this pattern relevant to "large datasets" — it's the same shape a real API call (GET /adventurers?page=2&pageSize=10) would take, just backed by an in-memory array with an artificial delay (SIMULATED_LATENCY_MS) instead of a network request.

Req 4 - Sorting and filtering [library/data/DataTablePage.tsx]

Sorting is wired the same way as pagination:

-grid owns the UI (clicking a column header).

-app owns the actual sort.

Code Explanation:

1. sortModel — which column is sorted and in which direction. It's an array because MUI's GridSortModel type supports multi-column sort, though this table only ever uses the first entry.
2. status — the current filter selection from the "Status" dropdown ("all", "hero", "villain", or "ally").
3. searchInput — the raw text currently typed into the search box, updated on every keystroke.
4. search — the "settled" search term that's actually sent to the API. It lags behind searchInput on purpose.

```
useEffect(() => {
  const timeout = setTimeout(() => setSearch(searchInput), 300);
  return () => clearTimeout(timeout);
}, [searchInput]);
```

Debounce: searchInput updates on every keystroke (so the text field feels instant), but search (the value actually sent to the fake API) updates 300ms after the user stops typing.

Without this, every keystroke would trigger a new fetch, which is wasteful and would cause visible flicker/race conditions as older requests resolve after newer ones.

Req 5 - Column Customization

Each column definition demonstrates a different customization option GridColDef supports:

- **headerName decouples the column's internal field key (adventurerClass) from what's shown to the user ("Class").**

Decouple: reducing the dependencies between different parts of a software system. Instead of having tightly connected components where changing one breaks another, this allows individual modules to be tested, updated, and maintained independently.

- **width vs flex — most columns get a fixed pixel width, but name uses flex: 1 so it grows to fill any remaining horizontal space, since names vary a lot in length and benefit from extra room.**
- **type: "number" tells the grid to right-align the column and sort/filter it numerically instead of as text.**
- **sortable: false on status disables the sort-by-click behavior for that column, since sorting by an arbitrary hero/villain/ally label wouldn't be meaningful.**
- **renderCell replaces the default plain-text cell with custom JSX — here, an MUI Chip whose label and color both come from lookup tables (statusLabels, statusColors) keyed by status, so each of the three statuses gets distinct, readable styling instead of plain text.**

Component Libraries (Advanced)

• Data display components (Table, List) [DONE]
• Feedback components (Alert, Snackbar, Dialog) [DONE]
• Navigation components (AppBar, Drawer, Tabs) [DONE]
• Form components (Select, Checkbox, Radio) [DONE]
• Customizing component styles [DONE]
Practice : Build a complete dashboard layout with library components [DONE]

Req 1 - Data display components (table, list) [ComponentsPage.tsx]

Lists: Ln109-130

Table: Ln154-173

It shows the pop up menu sidebar at library/components (drawer).

Req 2 - Feedback components (Alert, snackbar, dialog) [ComponentsPage.tsx]
Alert [Ln134]:
Snackbar [Ln230]: temporary UI component that displays brief messages after saving or resetting settings. It auto-dismisses after 2.5seconds.
Snackbar: handles the timing and position.
Alert within snackbar: supplies the colored message content.
Dialog [Ln241]: a floating UI window that appears temporarily on top of the main web page content. In my implementation, it's a blocking modal used specifically because resetting is destructive → it forces an explicit Cancel/Reset choice rather than happening instantly.

Req 3 - Navigation components (AppBar, Drawer, Tabs) [library/ComponentPages]
AppBar [Ln86]: main navigation or action bar. Typically positioned at the top of the screen displaying branding, current page title, navigation icons, and quick-action buttons (like search, settings, or user profiles).
Drawer [Ln107]: slide-in-side panel (opened by the menu icon) with links to other pages.

Tabs [Ln94]: which panel to render.

Req 4 - Form Components (Select, Checkbox, Radio) [library/ComponentPages]
Select [Ln194]: choose which difficulty you want- collapses after selection.
Checkbox [Ln213]: independent on/off toggle.
Radio [Ln204]: few, mutually-exclusive options shown all at once side-by-side (row), since there are only two (light/dark) and a dropdown would be overkill.
Q/Why did we use 3 components?
Many options → Select then collapse
Few visible options → Radio
One independent toggle → Checkbox

Req 5 - Customizing component styles
2 layers working together:

1. **Global theme (src/shared/mui/muiTheme.ts Ln56)**

Applies to every AppBar in the app automatically (That's why the one in ComponentPage.tsx never sets its own background color).

2. **Per-instance (features/library/ComponentsPage.tsx Ln86):**

Rounds the specific AppBar corners without affecting any other AppBar elsewhere.
Q/Why do we use both layers?
Global theme includes anything that should look consistent everywhere (defined once)

Per-instance includes small, situational tweaks meant for one specific spot on the page. That way both global theme and per instance stay contained to that single element instead of accidentally affecting every other place where that same component is used elsewhere in the app.

Practice: Build a complete dashboard layout with library components [ComponentsPage]
ComponentsPage.tsx is the answer. It's a one working page that combines navigation, data display, feedback, and forms.